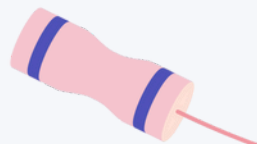


Recommendation System for Fitness and Nutrition

Supervised by Eng. Mahmoud Talaat





Done by Team members

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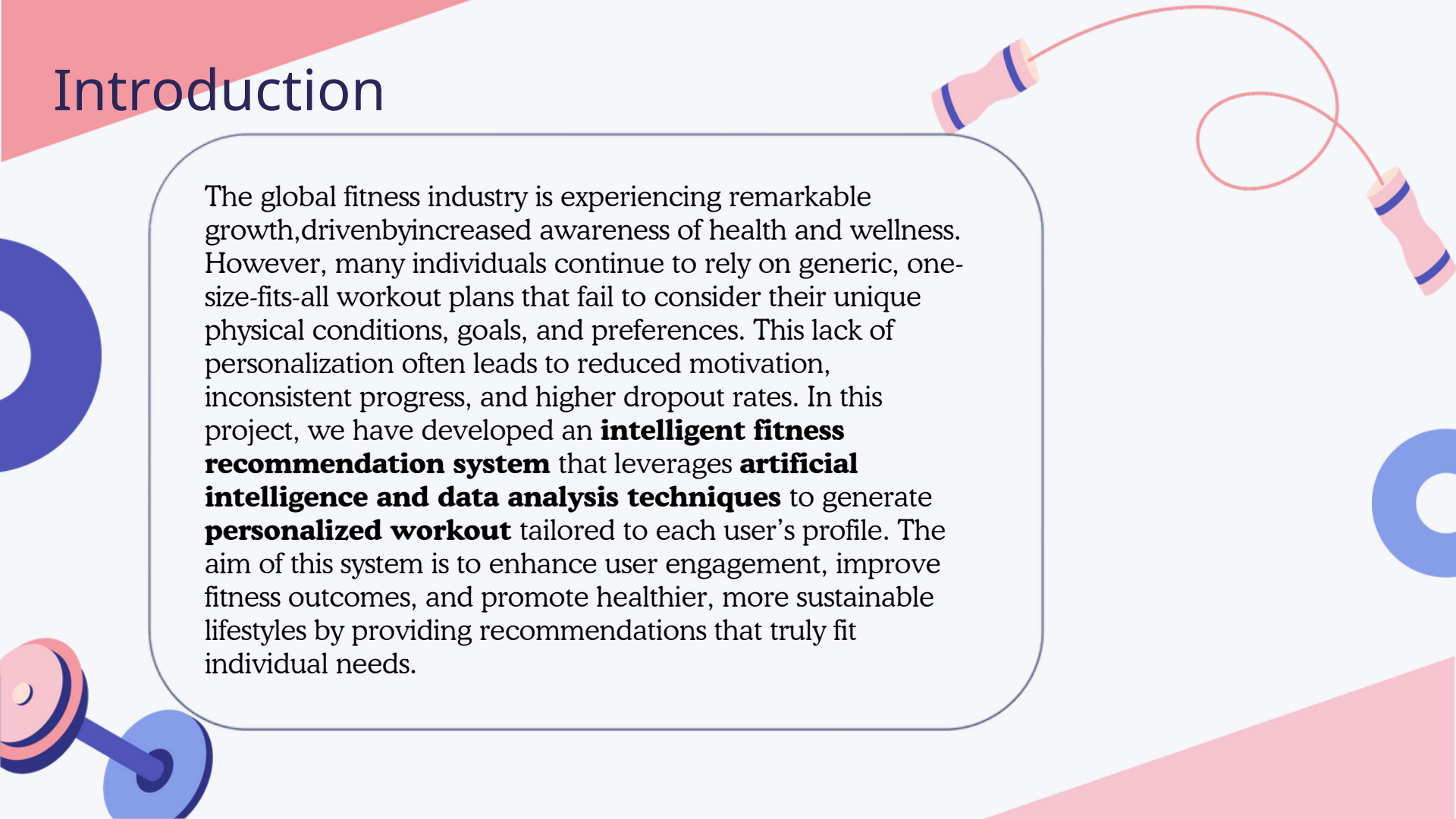
**Unsupervised Model
(phase 2)**

The background is a light blue-grey color. It features several abstract geometric elements: a red line forming a loop in the top-left; a pink and blue cylindrical object in the top-left; a blue square with the number '01' in the center; a blue cylindrical object in the top-right; a pink ring in the middle-right; a blue ring in the middle-left; a pink triangle in the bottom-left; and a blue and pink cylindrical object in the bottom-right.

01

Introduction

Introduction



The global fitness industry is experiencing remarkable growth, driven by increased awareness of health and wellness. However, many individuals continue to rely on generic, one-size-fits-all workout plans that fail to consider their unique physical conditions, goals, and preferences. This lack of personalization often leads to reduced motivation, inconsistent progress, and higher dropout rates. In this project, we have developed an **intelligent fitness recommendation system** that leverages **artificial intelligence and data analysis techniques** to generate **personalized workout** tailored to each user's profile. The aim of this system is to enhance user engagement, improve fitness outcomes, and promote healthier, more sustainable lifestyles by providing recommendations that truly fit individual needs.

The background is a light blue-grey color. It features several abstract geometric elements: a red line forming a loop in the top-left; a pink and blue striped cylinder in the top-left; a blue square with the number '02' in the center; a blue cylinder in the top-right; a pink ring in the middle-right; a blue ring in the middle-left; a pink triangle in the bottom-left; and a blue and orange striped cylinder in the bottom-right.

02

Problem statement

Current Challenges in Achieving Personalized Fitness

Despite the rapid expansion of the fitness industry, most individuals still struggle to achieve their fitness goals due to several real-world challenges:

- 🏋️ Generic workout plans fail to consider users' individual goals, physical abilities, or health conditions —leading to ineffective results and loss of motivation.
- 📉 Lack of professional guidance makes it difficult for beginners to design safe and efficient workout or nutrition routines.
- 📱 Limited personalization in existing fitness apps, which often recommend the same exercises regardless of user data or progress.
- 💡 Difficulty tracking and adapting to user progress—many systems do not update plans dynamically as users improve or face challenges



The background is a light gray with several abstract geometric elements. In the top left, there is a red line forming a loop and a pink cylinder with blue bands. In the top right, there is a blue cylinder. In the bottom right, there is a blue cylinder passing through a pink and blue striped sphere. On the left and right sides, there are partial views of blue and pink rings. The overall style is modern and minimalist.

03

Proposal solution

Intelligent Fitness Recommendation System

To address the limitations of traditional, one-size-fits-all workout approaches, we have developed an



AI-Based Fitness Recommendation System:

Our system provides personalized workout and nutrition guidance tailored to each user's unique profile. Users input key personal data — including age, weight, fitness goals, Experience level, and health conditions — and receive customized exercise routines designed specifically for them.

Using AI & data analytics, the system:



Analyzes user data to suggest the most suitable workouts and progression schedules.



Dynamically updates recommendations as users improve or face new challenges, ensuring continuous progress.

The goal is to make fitness training smarter, safer, and more effective by providing a personalized, data-driven solution that enhances motivation, consistency, and long-term health outcomes.

To overcome the limitations of traditional one-size-fits-all approaches, we developed a two-phase system:



Workout Type Prediction: Predicts the user's most suitable workout type from six classes.



Content-Based Recommendation: Generates a personalized workout routine based on the predicted type.

This approach ensures highly tailored, engaging, and results-oriented fitness guidance for every user.

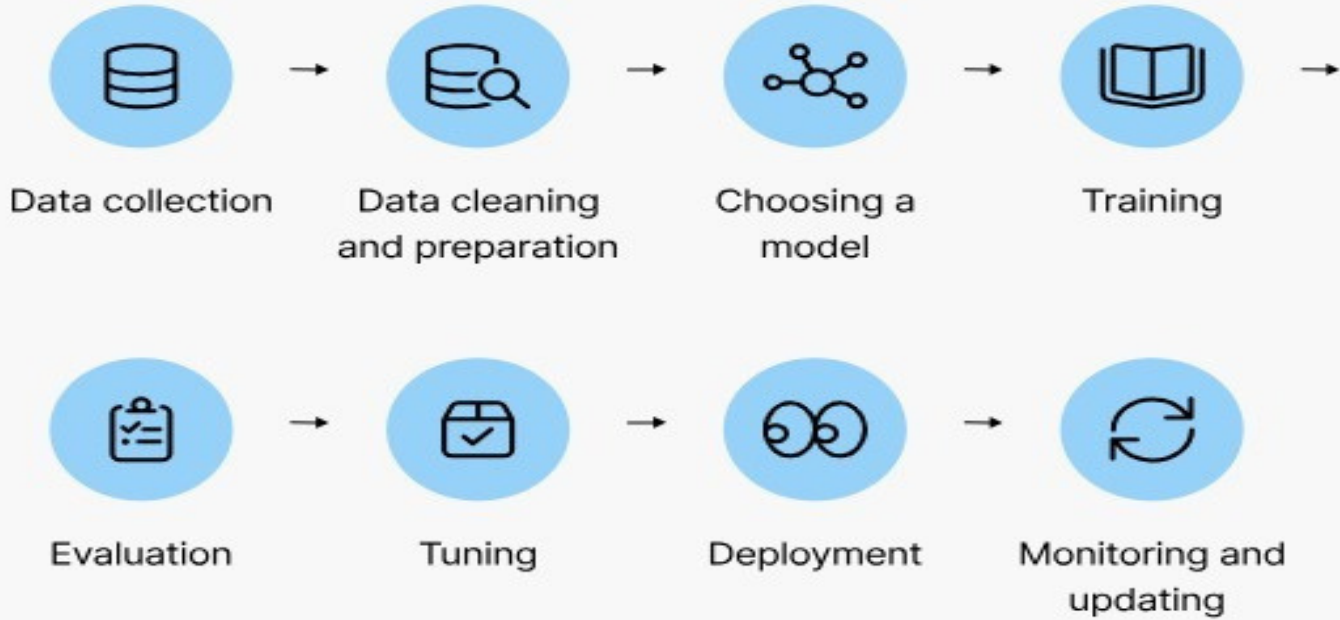


The background is a light blue-grey color. It features several abstract geometric elements: a red line forming a loop in the top-left; a pink and blue striped cylinder in the top-left; a blue square with the number '04' in the center; a blue cylinder in the top-right; a pink ring in the middle-right; a blue ring in the middle-left; a pink triangle in the bottom-left; and a blue and orange striped cylinder in the bottom-right.

04

Methodology

Methodology



Data collection

EXPLANATION FEATURES OF DATA: DATASET 1

- ❤️ Max_BPM: The maximum heart rate recorded during a workout session, indicating intensity level.
- 📶 Avg_BPM: The average heart rate throughout the session, used to measure effort and endurance.
- 😊 Resting_BPM: The user's resting heart rate, reflecting cardiovascular fitness and recovery level.
- ⌚ Session_Duration (hours): Indicates how long the workout session lasted, measured in hours.
- 🔥 Calories_Burned: The amount of energy (calories) burned during the workout session.
- 📅 Workout_Frequency (days/week): How many days per week the user typically exercises, used to gauge consistency.
- 🏋️ Experience_Level: Shows how advanced the user is (beginner(encoded to 1), intermediate(encoded to 2), advanced(encoded to 3)), guiding workout difficulty.
- 💧 Water_Intake (liters): The amount of water consumed daily in liters, used to assess hydration level.
- 🎯 Fitness_Goal: The user's main objective (weight loss, strength, endurance, muscle gain, etc.).
- 🏃 Workout_Type: The actual workout category performed (This is the ML prediction target.)
- 🏋️ Equipment_Used (One-Hot Encoded): indicates whether user have this equipment or not

Data collection

EXPLANATION FEATURES OF DATA: DATASET 2

-  bodyPart: The primary body part targeted by the exercise (e.g., Chest, Back, Legs).
-  equipment: The equipment used for the exercise (e.g., Dumbbell, Barbell, Bodyweight).
- id: Unique identifier for each exercise in the dataset.
-  name: The name of the exercise (e.g., "Push-Up", "Squat").
-  target: The main muscle group worked by the exercise (e.g., Biceps, Quads, Chest).
-  Secondary Muscles: Combines all secondary muscles (secondaryMuscles/0.../5) into a single column, showing additional muscles engaged by the exercise. Example: "Biceps, Forearms, Shoulders".
-  Instructions (Full): Combines all instruction steps (instructions/0.../10) into a single column providing the complete step-by-step guide for performing the exercise.

Data cleaning

for **users dataset** we replaced Nan values with median since it's robust to outliers and we filled missing values with 0 for all equipment-related columns

```
df['Fat_Percentage'].fillna(df['Fat_Percentage'].median(), inplace=True)
df['Water_Intake (liters)'].fillna(df['Water_Intake (liters)'].median(), inplace=True)

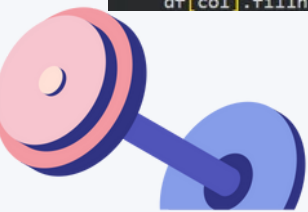
equipment_cols = ['Equipment_dumbbell', 'Equipment_barbell', 'Equipment_cable',
                  'Equipment_kettlebell', 'Equipment_resistance_band',
                  'Equipment_stability_ball', 'Equipment_bodyweight_only']

for col in equipment_cols:
    df[col].fillna(0, inplace=True)
```

```
users.isna().sum()
```

Columns with missing values:

Fat_Percentage	120
Water_Intake (liters)	80
Equipment_dumbbell	190
Equipment_barbell	190
Equipment_cable	190
Equipment_kettlebell	190
Equipment_resistance_band	190
Equipment_stability_ball	190
Equipment_bodyweight_only	190



Data analysis

In data analysis, we use key commands to understand the dataset before training. `users.info()` provides basic information about the columns, such as data types, non-null values, and overall structure. `users.nunique()` shows the number of unique values in each column, helping us identify whether a feature is useful for the model. Meanwhile, `users.describe()` gives a statistical summary of numerical columns, including the mean, median, standard deviation, and range, which helps us understand data distribution and detect outliers. All of this is essential for properly preparing the data before building a machine learning model.

```
users_df.nunique()
```

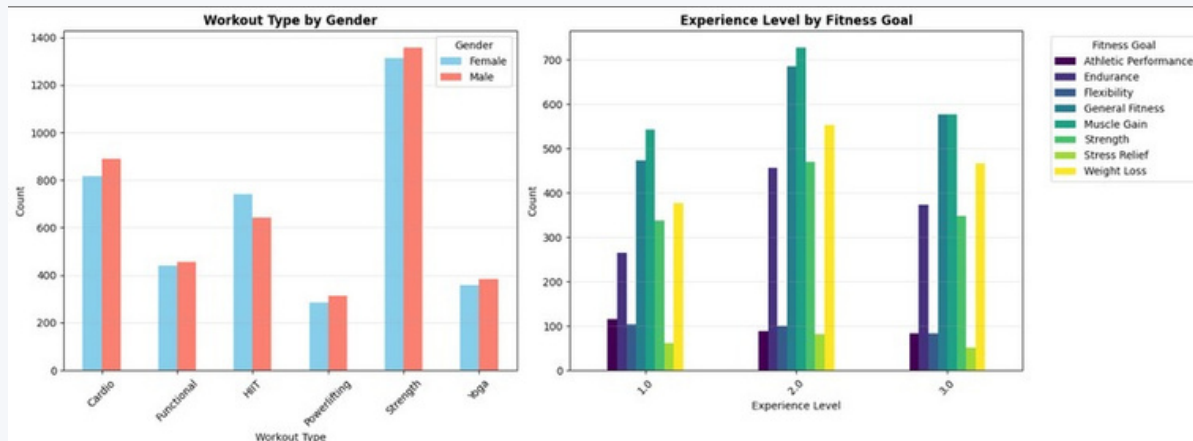
	count	mean	std	min	25%	50%	75%	max
Age	8000.0	34.723512	12.052514	16.00	25.7750	32.60	42.20	72.50
Weight (kg)	8000.0	68.868425	15.072511	40.00	58.6000	67.70	77.90	137.20
Height (m)	8000.0	1.706734	0.110925	1.45	1.6300	1.70	1.78	2.05
BMI	8000.0	23.739225	5.106056	10.00	20.6975	23.66	26.59	59.21
Fat_Percentage	7880.0	20.753680	5.780239	7.60	16.5000	20.60	24.70	45.00
Max_BPM	8000.0	184.169750	13.603001	140.00	175.0000	185.70	194.20	210.00
Avg_BPM	8000.0	141.818025	21.342400	90.00	126.5000	142.40	158.00	197.00
Resting_BPM	8000.0	63.043563	7.949840	45.00	57.1000	63.30	69.20	85.20
Session_Duration (hours)	8000.0	1.228191	0.436143	0.25	0.9000	1.19	1.51	2.50
Calories_Burned	8000.0	736.219300	371.992761	50.00	464.5750	694.15	979.00	2000.00
Workout_Frequency (days/week)	8000.0	3.737250	0.965231	1.00	3.0000	3.80	4.40	6.10
Experience_Level	8000.0	2.035250	0.776906	1.00	1.0000	2.00	3.00	3.00
Water_Intake (liters)	7920.0	2.943884	0.621191	1.50	2.5000	3.00	3.40	5.00
Equipment_dumbbell	7810.0	0.552113	0.497309	0.00	0.0000	1.00	1.00	1.00
Equipment_barbell	7810.0	0.459923	0.498423	0.00	0.0000	0.00	1.00	1.00
Equipment_cable	7810.0	0.424328	0.494272	0.00	0.0000	0.00	1.00	1.00
Equipment_kettlebell	7810.0	0.422151	0.493934	0.00	0.0000	0.00	1.00	1.00
Equipment_resistance_band	7810.0	0.389685	0.487755	0.00	0.0000	0.00	1.00	1.00
Equipment_stability_ball	7810.0	0.300256	0.458399	0.00	0.0000	0.00	1.00	1.00
Equipment_bodyweight_only	7810.0	0.748784	0.433740	0.00	0.0000	1.00	1.00	1.00

Data columns (total 23 columns):			
#	Column	Non-Null Count	Dtype
0	Age	8000 non-null	float64
1	Gender	8000 non-null	object
2	Weight (kg)	8000 non-null	float64
3	Height (m)	8000 non-null	float64
4	BMI	8000 non-null	float64
5	Fat_Percentage	7880 non-null	float64
6	Max_BPM	8000 non-null	float64
7	Avg_BPM	8000 non-null	float64
8	Resting_BPM	8000 non-null	float64
9	Session_Duration (hours)	8000 non-null	float64
10	Calories_Burned	8000 non-null	float64
11	Workout_Frequency (days/week)	8000 non-null	float64
12	Experience_Level	8000 non-null	float64
13	Water_Intake (liters)	7920 non-null	float64
14	Fitness_Goal	8000 non-null	object
15	Workout_Type	8000 non-null	object
16	Equipment_dumbbell	7810 non-null	float64
17	Equipment_barbell	7810 non-null	float64
18	Equipment_cable	7810 non-null	float64
19	Equipment_kettlebell	7810 non-null	float64
20	Equipment_resistance_band	7810 non-null	float64
21	Equipment_stability_ball	7810 non-null	float64
22	Equipment_bodyweight_only	7810 non-null	float64

0	bodyPart	1324 non-null	object
1	equipment	1324 non-null	object
2	gifUrl	1324 non-null	object
3	id	1324 non-null	int64
4	name	1324 non-null	object
5	target	1324 non-null	object
6	secondaryMuscles/0	1324 non-null	object
7	secondaryMuscles/1	986 non-null	object
8	instructions/0	1324 non-null	object
9	instructions/1	1324 non-null	object
10	instructions/2	1324 non-null	object
11	instructions/3	1324 non-null	object
12	instructions/4	1242 non-null	object
13	instructions/5	739 non-null	object
14	secondaryMuscles/2	233 non-null	object
15	instructions/6	313 non-null	object
16	instructions/7	92 non-null	object
17	secondaryMuscles/3	32 non-null	object
18	instructions/8	20 non-null	object
19	secondaryMuscles/4	4 non-null	object
20	instructions/9	5 non-null	object
21	secondaryMuscles/5	2 non-null	object
22	instructions/10	3 non-null	object



Users Dataset



Workout Type by Gender

- Most Popular: Strength training is the clear favorite for both genders (~1,330 females, ~1,370 males).
- Gender Similarities: Participation is fairly balanced across most workout types.
- Least Popular: Powerlifting (~290 females, ~310 males) is the least chosen.
- Other Trends: Cardio is second (~820 females, ~900 males), HIIT third (~740 females, ~640 males).

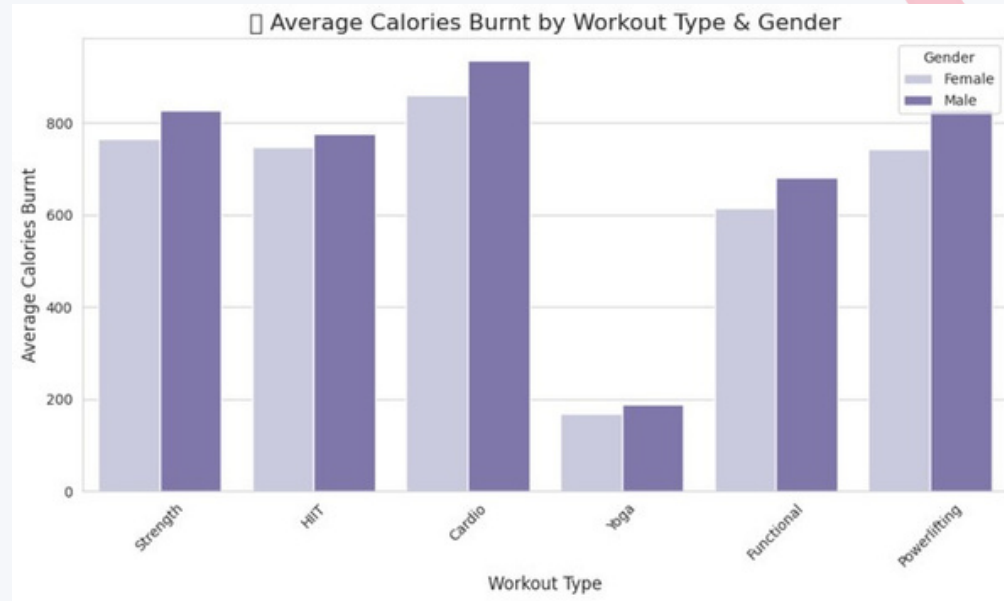
Experience Level by Fitness Goal

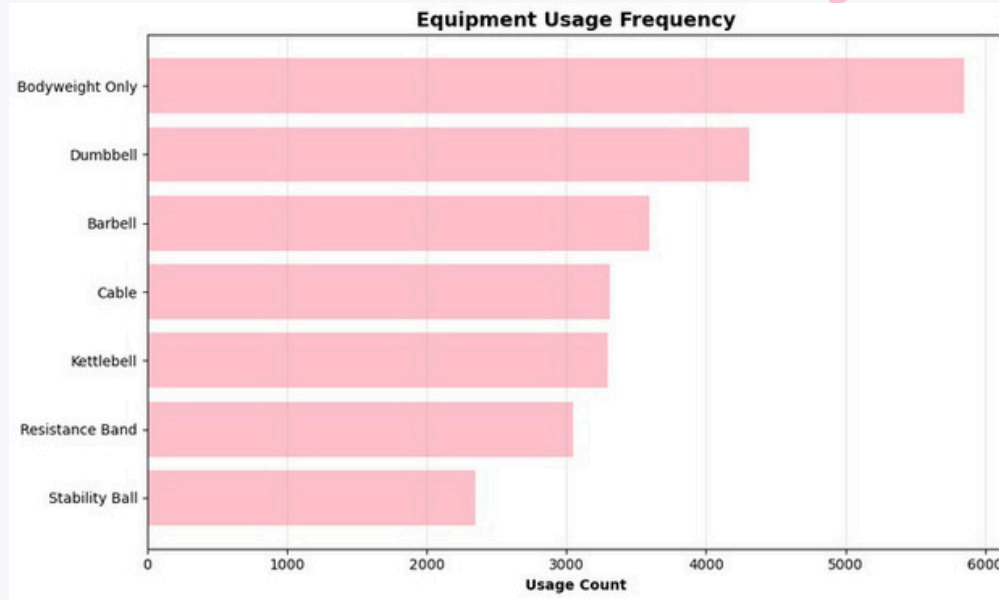
- General Fitness Leads: Most common goal across all levels, peaking at intermediate (2.0) with 700+ participants.
- Experience Distribution: Intermediate level (2.0) has the highest participation overall.
- Weight Loss: Popular among beginners (~380) and intermediates (~550), lower at advanced (~470).
- Specialized Goals: Athletic Performance and Flexibility remain niche across all levels.
- Advanced Focus: Muscle Gain rises as the second goal at advanced level (3.0, ~580), showing targeted strength focus.



Calories Burned by Workout Type and Gender:

- 👤 Gender: Males burn more calories than females; gap largest in Cardio, smallest in Yoga.
- 🔥 Top Workout: Cardio — highest calorie burn (M: ~930 kcal, F: ~870 kcal).
- 🧘 Lowest Workout: Yoga — lowest calories (M: ~190 kcal, F: ~165 kcal).
- 💪 Mid-Range: Strength, HIIT, Powerlifting → 750–830 kcal.
- ✅ Takeaway: Cardio = max calorie burn; Resistance training = good calorie burn + muscle-building.

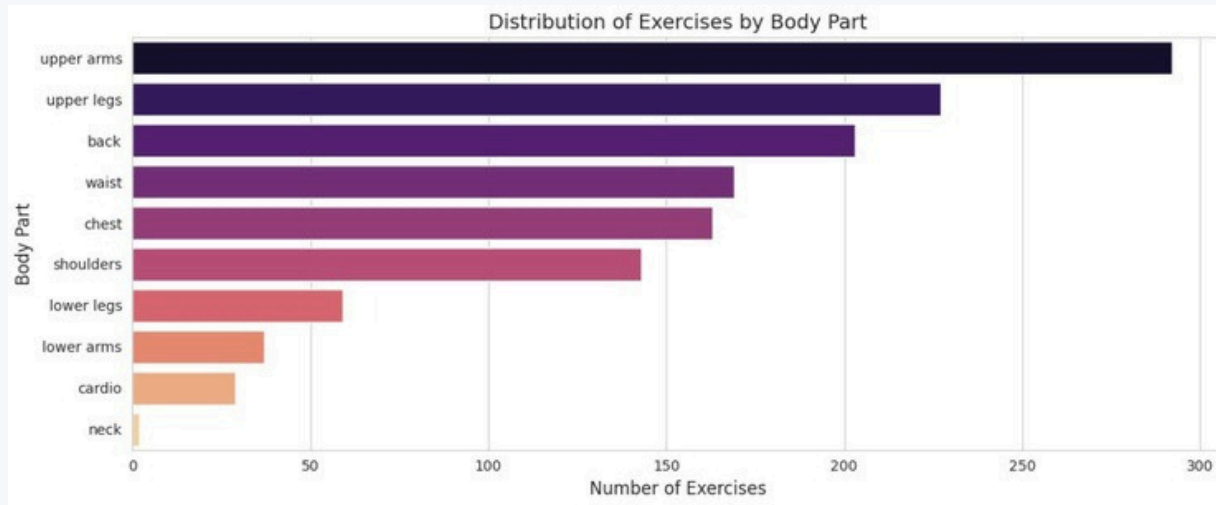




Equipment Popularity Rankings:

- Most Popular: Bodyweight exercises dominate (~5,850 uses), showing preference for accessible, no-equipment workouts.
- Top Free Weights: Dumbbells (~4,300) and barbells (~3,600) follow, highlighting traditional strength training's importance.
- Mid-Tier: Cable machines, kettlebells, and resistance bands (~3,000–3,300 uses) show moderate, similar popularity.
- Least Used: Stability balls (~2,350 uses) are far less common.

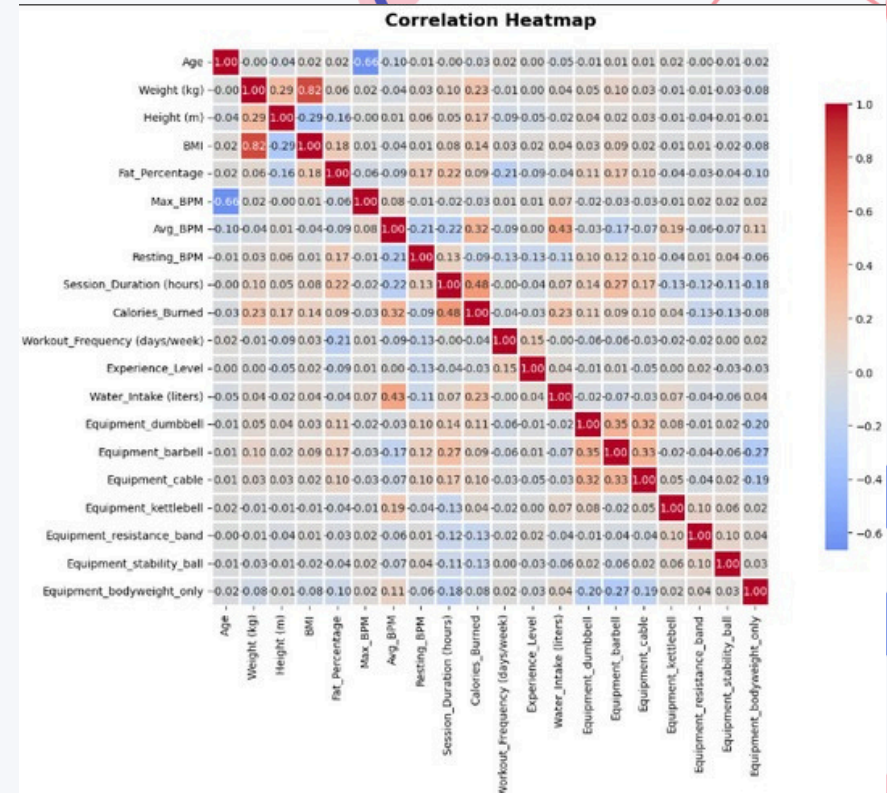
Exercises Dataset



- **Dataset Focus:** The dataset is overwhelmingly focused on strength training, particularly for the upper body.
- **Highest Exercise Variety:** "Upper arms" has the most exercise variety by a wide margin (nearly 300 exercises), followed by "upper legs" (approx. 235) and "back" (approx. 210).
- **Moderate Variety:** The "waist" (core), "chest", and "shoulders" have a solid, moderate number of exercises available (ranging from ~145 to 165).
- **Significant Gaps:** There is a sharp drop-off in exercise availability for "lower legs" (~60), "lower arms" (~40), and cardio (~30).
- **Negligible Category:** "Neck" exercises are almost non-existent in this dataset.

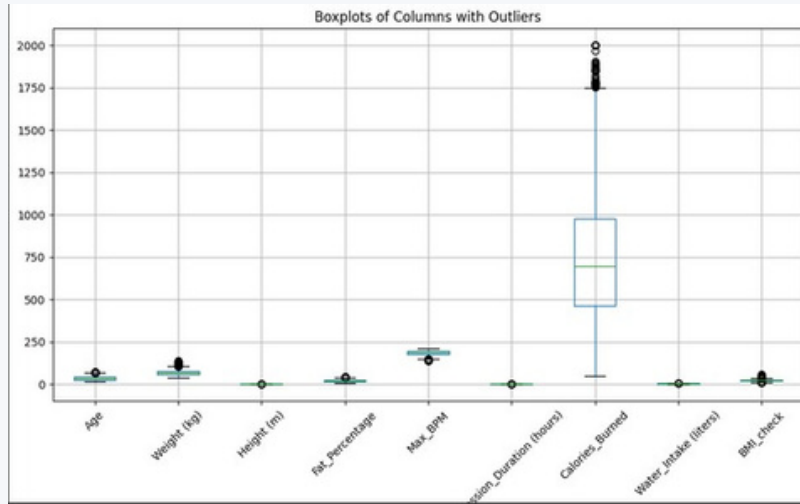
Correlation Analysis:

- 💪 Physical & Heart Metrics: Weight ↔ BMI (0.92), Max_BPM ↔ Avg_BPM (0.86)
- 🏋️ Equipment Patterns: Dumbbell ↔ Barbell (0.35), Resistance Band ↔ Stability Ball (0.32)
- 📉 Negative Trends: Longer sessions → lower body fat (-0.25); slight cardio-strength trade-offs
- 😐 Weak/No Correlations: Age, Workout Frequency, Experience Level mostly independent
- 🔥 Takeaway: Fitness outcomes are multifactorial → focus on multiple factors rather than one metric

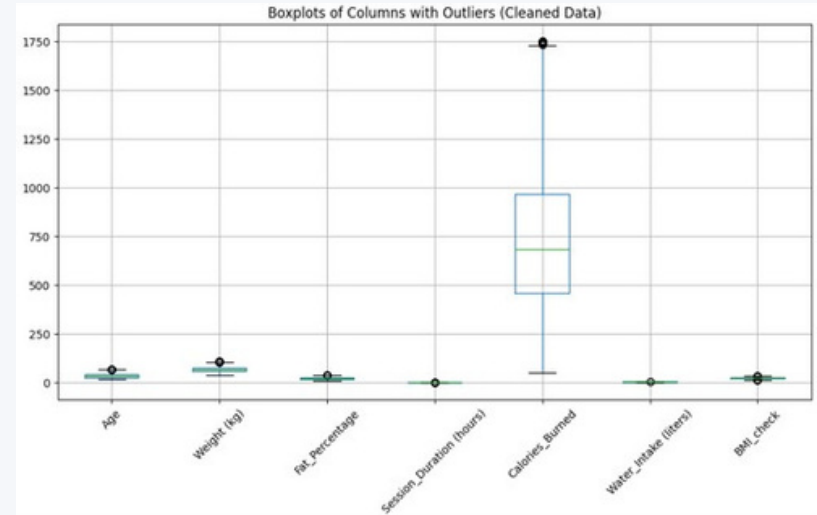


Outlier Detection & Handling

before



after



we kept just the outliers that make sense



We created 10 new features to enhance the fitness dataset, focusing on performance, efficiency, and body composition. Key metrics include Intensity Score and HR Reserve/HR Intensity for cardiovascular performance, Calorie per kg and Session Intensity for workout efficiency, and a Body Composition Score combining BMI and body fat percentage to reflect overall fitness

[illegible]

Exercise Data Preprocessing & Feature Engineering

Data Cleaning & Consolidation:

- Combined multiple secondary muscle columns into a single secondary_muscles column.
- Merged multiple instruction columns into instructions_full for complete steps and instructions_short for the first 3 steps.

Enhanced Muscle Group Mapping

- Standardized target muscles into broader groups (e.g., Chest, Back, Shoulders, Core).
- Categorized muscles by size: Big (Chest, Back, Quads, Hamstrings) vs Small (Shoulders, Biceps, Triceps, etc.).

Equipment Difficulty & Encoding

- Assigned a difficulty score based on equipment type (Body weight = Easy, Dumbbells = Moderate, Barbells = Hard).
- Encoded categorical features (equipment and muscle_group) for modeling using label encoding.

```
difficulty_mapping = {
    'body weight': 1, 'assisted': 1, 'band': 1, 'resistance band': 1,
    'dumbbell': 2, 'kettlebell': 2, 'medicine ball': 2, 'stability ball': 2,
    'cable': 2, 'leverage machine': 2, 'rope': 2, 'bosu ball': 2,
    'barbell': 3, 'ez barbell': 3, 'olympic barbell': 3, 'smith machine': 3,
    'trap bar': 3, 'sled machine': 3, 'weighted': 3
}

clean_df['difficulty_level'] = clean_df['equipment'].map(difficulty_mapping).fillna(2)

# Exercise type mapping
exercise_type_mapping = {
    'waist': 'Core', 'back': 'Strength', 'chest': 'Strength',
    'upper arms': 'Strength', 'shoulders': 'Strength', 'upper legs': 'Strength',
    'lower legs': 'Strength', 'lower arms': 'Strength', 'cardio': 'Cardio',
    'neck': 'Flexibility'
}

clean_df['exercise_type'] = clean_df['bodyPart'].map(exercise_type_mapping)

# Accessibility score (1=gym only, 5=home friendly)
accessibility_mapping = {
    'body weight': 5, 'band': 5, 'resistance band': 5,
    'dumbbell': 4, 'kettlebell': 4, 'medicine ball': 4,
    'stability ball': 3, 'cable': 2, 'barbell': 2,
    'smith machine': 1, 'sled machine': 1, 'leverage machine': 1
}

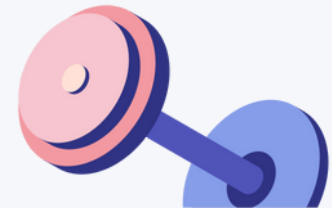
clean_df['accessibility'] = clean_df['equipment'].map(accessibility_mapping).fillna(3)

# Calculate complexity score based on instruction length
clean_df['complexity'] = clean_df['instructions'].str.len() / 100
clean_df['complexity'] = clean_df['complexity'].clip(1, 5)

return clean_df

# Clean exercise dataset
exercises = clean_exercises_dataset(exercises_raw)
print(f"Exercise dataset cleaned: {exercises.shape}")

# Display basic statistics
print("\nDATASET OVERVIEW")
print("-" * 30)
print("User Dataset Features:")
```



Exercise Data Preprocessing & Feature Engineering

Feature Vector Creation

- Each exercise was converted into a structured feature vector including: bodyPart, target, equipment, difficulty_score, and muscle_group.
- Ensures each exercise can be compared numerically for similarity computation.

Categorical Feature Encoding

- One-hot encoded categorical features (body Part, target, equipment, muscle_group).
- Combined with numeric difficulty_score to create a unified feature matrix.
- Standardized numeric features (0-1 range) for consistent similarity calculations.

```
# Code by Harshvardhan P. Kulkarni - @HarshvardhanP
# Exercise Data Preprocessing

# Combines secondary muscles, creates full + short instructions

# 1. Copy original DataFrame
exercises_clean = exercises_df.copy()

# 2. Combine secondary muscles into one column
secondary_cols = [col for col in exercises_clean.columns if 'secondaryMuscles' in col]
exercises_clean['secondary_muscles'] = exercises_clean[secondary_cols].apply(
    lambda row: ' '.join([str(val) for val in row if pd.notna(val) and val != 'NaN']),
    axis=1
)

# 3. Combine ALL instruction steps into 'instructions_full'
instruction_cols = [col for col in exercises_clean.columns if 'instructions' in col]
exercises_clean['instructions_full'] = exercises_clean[instruction_cols].apply(
    lambda row: ' '.join([str(val) for val in row if pd.notna(val) and val != 'NaN']),
    axis=1
)

# 4. Create 'instructions_short' with first 3 steps
def get_first_3_instructions(row):
    steps = []
    for i in range(3):
        col = f'instructions_{i}'
        if col in row.index and pd.notna(row[col]) and row[col] != 'NaN':
            steps.append(f'{i+1}. {row[col]}')
    return "\n".join(steps) if steps else "No instructions available"

exercises_clean['instructions_short'] = exercises_clean.apply(get_first_3_instructions, axis=1)

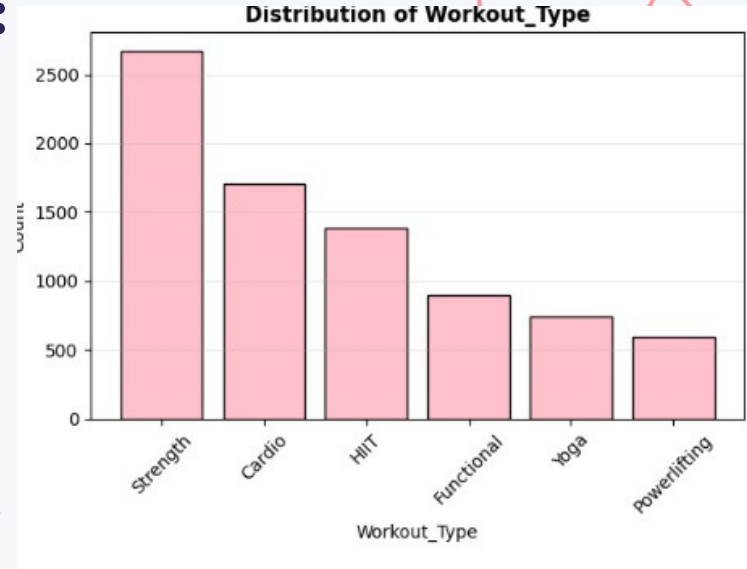
# 5. Keep only relevant columns (plus gifUrl)
exercises_clean = exercises_clean[[
    'id', 'name', 'bodyPart', 'target', 'equipment',
    'secondary_muscles', 'instructions_full', 'instructions_short'
]]

# 6. Verify results
print("Preprocessed exercises:", exercises_clean.shape)
print("Sample full instructions:\n", exercises_clean['instructions_full'].iloc[0])
print("Sample short instructions:\n", exercises_clean['instructions_short'].iloc[0])
```



Workout Type Distribution(Target):

- **Strength** is the most recorded workout, dominating the dataset.
- **Cardio and HIIT** follow, showing strong user interest in endurance and high-intensity training.
- **Functional workouts** appear at a moderate level.
- **Yoga and Powerlifting** are the least frequent categories.
- **Overall**, the data shows clear class imbalance, which must be handled during modeling to avoid biased predictions.



imbalanced Data Handling

During data exploration, we observed that the distribution of fitness categories—such as workout types or nutrition levels—was highly imbalanced. Some categories contained significantly more samples than others. This imbalance can cause the model to favor majority classes, leading to biased predictions and lower accuracy for underrepresented groups.

Techniques Used to Address Imbalance

- ✓ SMOTE (Synthetic Minority Oversampling Technique):
 - Generates new, synthetic samples for minority classes rather than simply duplicating existing ones.
 - Creates these samples by interpolating between nearest neighbors, helping the model learn smoother decision boundaries.
 - Ensures that all fitness categories are represented more evenly in the training data, improving fairness and model performance

```
📊 Class distribution BEFORE SMOTE:  
Cardio: 1637 (21.2%)  
Functional: 892 (11.5%)  
HIIT: 1334 (17.2%)  
Powerlifting: 599 (7.7%)  
Strength: 2542 (32.8%)  
Yoga: 735 (9.5%)
```

```
📊 Class distribution AFTER SMOTE:  
Cardio: 2033 (26.3%)  
Functional: 2033 (26.3%)  
HIIT: 2033 (26.3%)  
Powerlifting: 2033 (26.3%)  
Strength: 2033 (26.3%)  
Yoga: 2033 (26.3%)
```



The background is a light blue gradient. It features several abstract geometric elements: a red string forming a loop in the top left; a pink and blue cylindrical object in the top left; a blue cylindrical object in the top right; a pink ring on the right; a blue ring on the left; and a pink and blue cylindrical object in the bottom right.

05

Machine Learning Results (phase one)

Models Training

To find the best model for predicting workout types, we tried several tree-based classifiers:

- ⚡ Gradient Boosting
- 🌲 Random Forest
- 🌳 Extra Trees
- 🚀 HistGradientBoosting

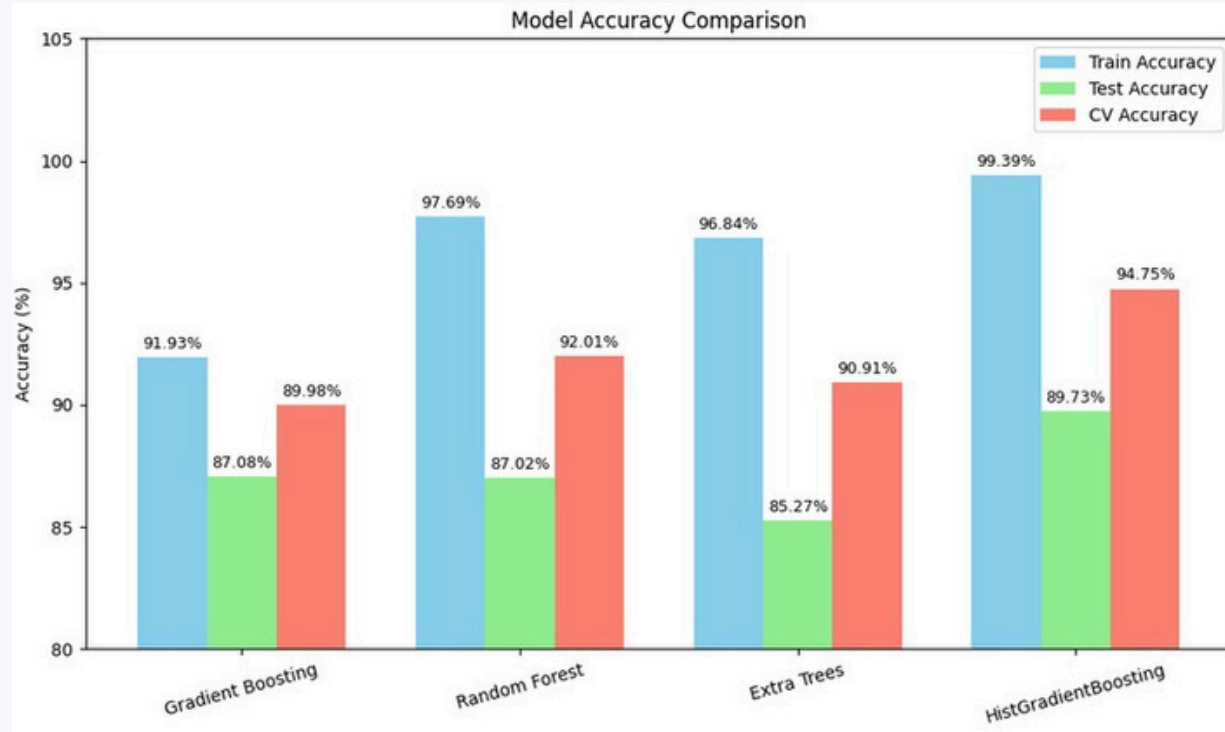
Each model was trained on the same data, and we evaluated them using:

- ✅ Train Accuracy
- 📊 Test Accuracy
- 🔄 5-Fold Cross-Validation Accuracy



MODELS COMPARISON

- HistGradientBoosting demonstrates the best generalization (highest CV score)
- All models show some overfitting (train > test), but within acceptable ranges
- HistGradientBoosting achieves the best bias-variance tradeoff
- Test accuracies range from 85-90%, indicating robust real-world performance



Model Selection

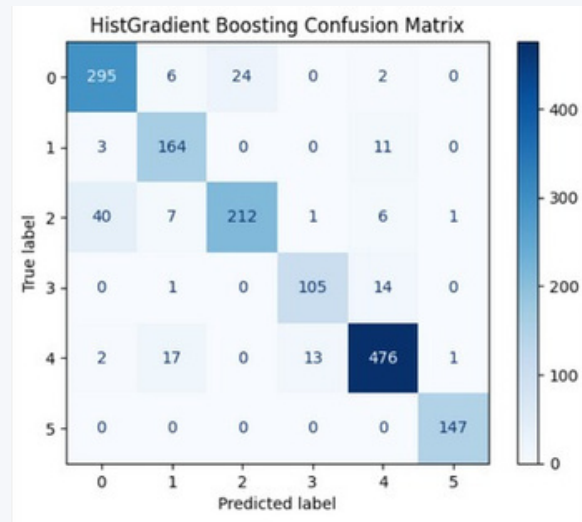


Why We Selected HistGradientBoosting?

Based on our comprehensive model evaluation, HistGradientBoosting was selected as the optimal model for predicting fitness experience levels due to its superior performance across all key metrics:

CONFUSION MATRIX:

- the model correctly classifies most samples across all 6 classes, showing strong performance overall. ✓
- Classes 4 and 5** have the highest accuracy with very few errors.
- Classes 0 and 1** also perform well with minimal misclassifications.
- Class 2 and Class 3** show moderate confusion with neighboring classes, especially 0, 1, and 4.
- Misclassifications** are relatively small compared to correct predictions, meaning the model generalizes well despite class imbalance.



Feature Importance

Top Predictors of Workouttype:

Fitness_Goal (~19%)

- Most influential feature
- User goals (weight loss, muscle gain, endurance) strongly indicate experience
- Shuffling causes largest performance drop

Avg_BPM (~8%)

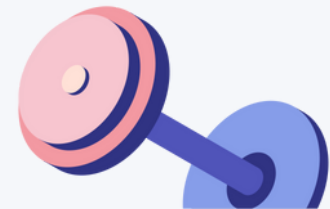
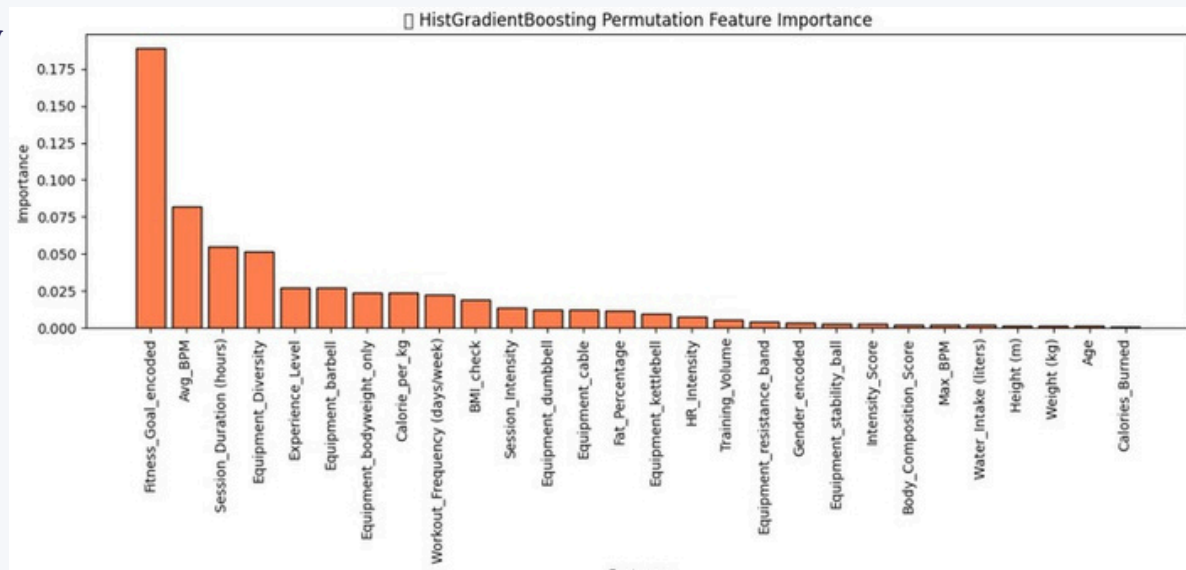
- Average heart rate during workouts
- Reflects cardiovascular fitness and workout intensity

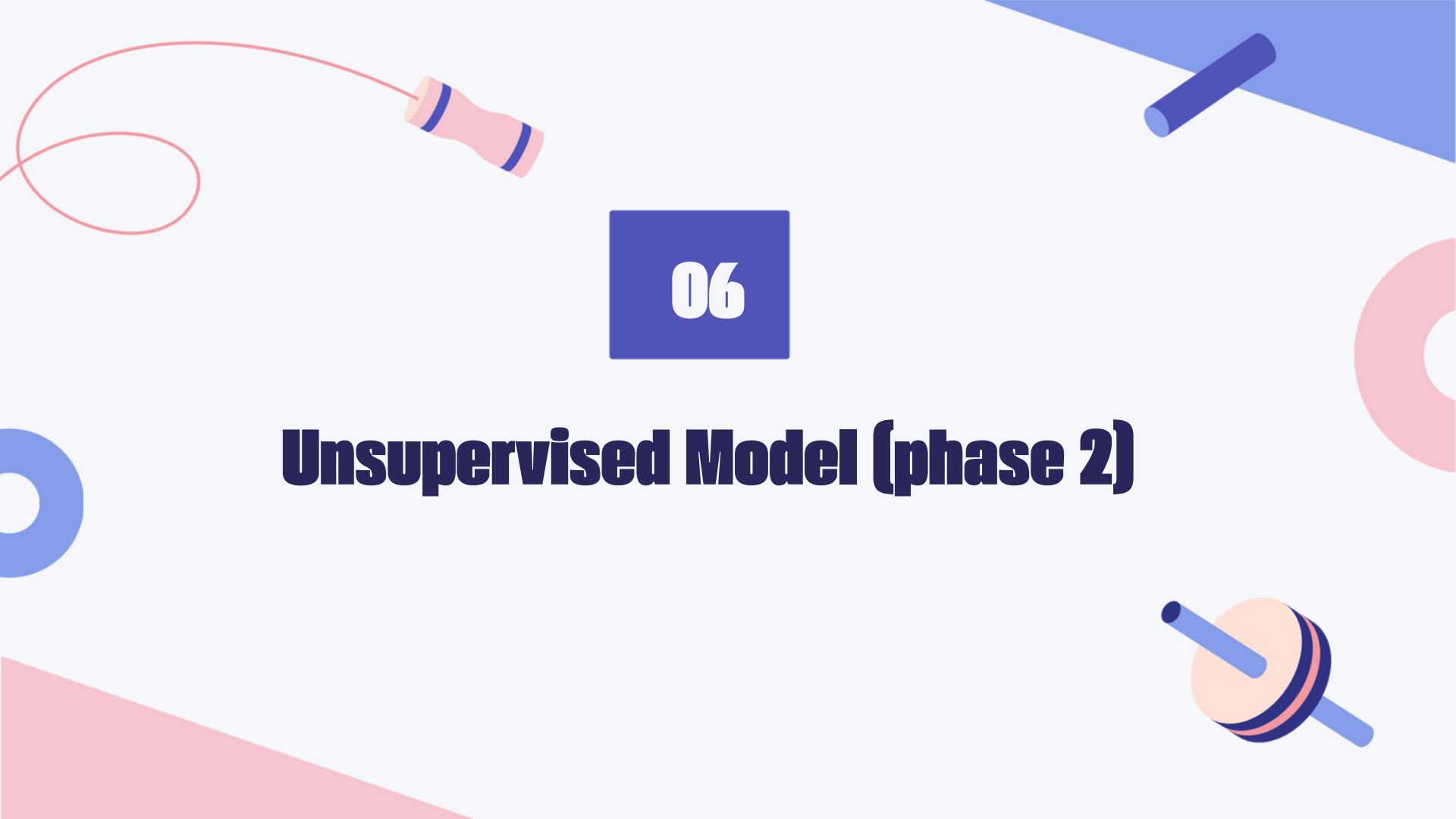
Session_Duration (hours) (~6%)

- Longer training sessions usually indicate higher experience
- Captures commitment and patterns of users

Equipment_Diversity (~5%)

- Variety of equipment used shows adaptability and



The background is a light gray with several abstract geometric elements. In the top left, a red line forms a loop, ending in a pink and blue cylindrical object. In the top right, a blue cylinder lies horizontally. On the right side, a pink ring is partially visible. In the bottom right, a blue cylinder passes through a stack of orange and blue rings. A large pink triangle is in the bottom left corner, and a blue triangle is in the top right corner.

06

Unsupervised Model (phase 2)

Content-Based Filtering pipeline

Content-Based Filtering(ExerciseRecommendation)

Input:User Constraints (injuries, equipment and experience level)



FeatureVectorization(1,324 exercises $\times$ 72 features each)



CosineSimilarityMatrix (1,324 $\times$ 1,324)



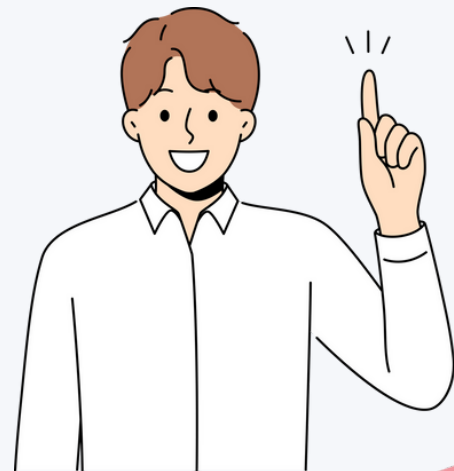
InjuryFiltering(removes unsafe exercises)



Output:Top N exercises per muscle group



DiversityFilter(ensures variety - not 3 curl variations)



Content-Based Filtering pipeline



Personalization Engine

Input: Predicted workout type + User goals/injuries



Generate N-days split plan with sets/reps based on goal



Calculate macros (Mifflin-St Jeor formula)



Sleep recommendations (based on training volume)



Output: Complete workout plan with nutrition



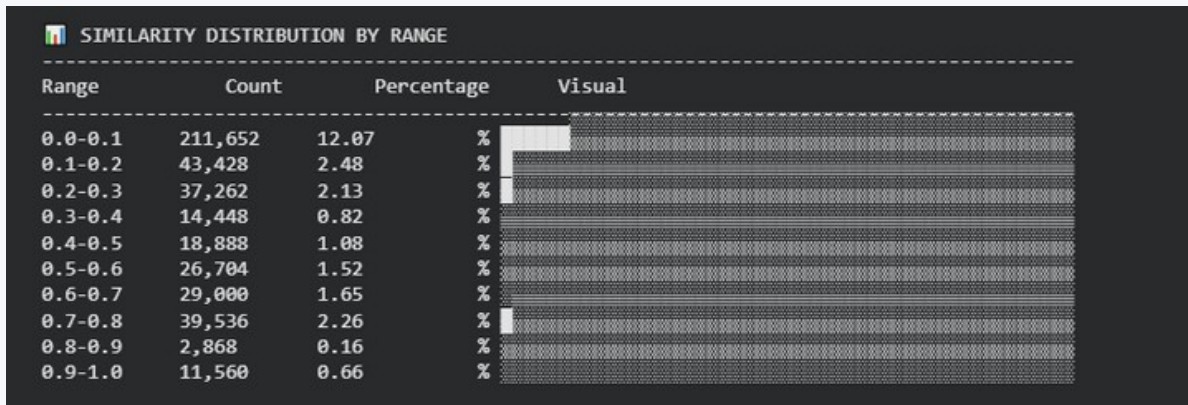
CONTENT-BASED FILTERING

How does it work?

Each exercise is converted to a 72-dimensional feature vector

- Each categorical feature gets one-hot encoded (turns into 0s and 1s)
- One-hot encoding = dumbbell bench press becomes (1,0,0,...,0) for dumbbell, (1,0,0,...,0) for barbell category, etc.
- Captures all exercise attributes in a format computers can compare

We compute a similarity matrix using cosine similarity



CONTENT-BASED FILTERING



Cosine Similarity:

What does this number mean?

- 1.0 = identical exercises (same muscle, equipment, difficulty)
- 0.65 = very similar (dumbbell bench press vs machine chest press)
- 0.0 = completely different muscle groups
- -0.23 = opposite characteristics (rare)

Key Insights:

- 87.53% of similarities are NEGATIVE or near-zero → exercises are diverse
- Only 4.07% have similarity > 0.6 → true duplicates are rare
- We can use 0.65 threshold to recommend similar yet diverse exercises

Unsupervised Pipeline



automatic target muscle group assignment

- INPUT: User specifies training frequency (3/4/5 days/week)



- SYSTEM AUTO-GENERATES MUSCLE SPLIT

```
workoutsplitmap = {  
  3: {  
    1: ["Chest", "Triceps"],  
    2: ["Back", "Biceps"],  
    3: ["Quads", "Hamstrings", "Calves", "Glutes", "Core"],  
  },  
  4: {  
    1: ["Chest", "Triceps"],  
    2: ["Back", "Biceps"],  
    3: ["Quads", "Hamstrings", "Calves"],  
    4: ["Shoulders", "Forearms"],  
  },  
  5: {  
    1: ["Chest", "Triceps"],  
    2: ["Back", "Biceps"],  
    3: ["Quads", "Hamstrings", "Calves"],  
    4: ["Shoulders", "Forearms"],  
    5: ["Glutes", "Core"],  
  },  
}
```

CONTENT-BASED FILTERING

Equipment Filtering :

Filter by equipment available:

- If Gym: Keep all (barbell, dumbbell, machine, cable, etc.)

- If Home: equipment_map = {

- 'a': ['body weight', 'band'], #Minimal (body weight, resistance band only)

- 'b': ['body weight', 'band', 'dumbbell'], #Basic (+ dumbbell)

- 'c': ['body weight', 'band', 'dumbbell', 'kettlebell', 'medicine ball', 'stability ball']

- , #Standard (+ kettlebell, medicine ball, stability ball)

- 'd': ['body weight', 'band', 'dumbbell', 'kettlebell', 'medicine ball', 'stability ball', 'cable']

-], #Full home setup

- }

- If Both : Keep all

- 144 exercises remain



Difficulty Filtering (Experience Match):

67 chest exercises (after equipment filter)

Match difficulty to experience:

- Beginner (Level 1): Only difficulty 1 exercises

- Intermediate (Level 2): Difficulty 1-2 exercises

- Advanced (Level 3): All difficulties 1-3

Example (Intermediate): ----> Keep exercises with difficulty ≤ 2

Injury Handler (CRITICAL SAFETY STEP)



Injury Filtering :

After filtering by muscle, equipment, and difficulty, we still have exercises that could harm users with injuries.

```
INPUT:
├ Exercise pool (e.g., 94 safe chest exercises)
├ User injuries (e.g., ["shoulder injury", "lower back pain"])
└ INJURY_KEYWORDS dictionary
```

PROCESS:

```
for injury in user_injuries:
    keywords = INJURY_KEYWORDS[injury]

    for exercise in exercise_pool:
        # Check BOTH exercise name AND target muscle
        if any(keyword in exercise.name) OR
           any(keyword in exercise.target):
```

```
        X REMOVE exercise from pool
```

Example:

```
└ "Overhead Press" contains "overhead"
  (matches shoulder keyword)
  → REMOVED
```

```
return filtered_safe_exercises
```

Starting pool: 144 chest exercises



Keyword matching for "shoulder", "overhead", "press", etc.

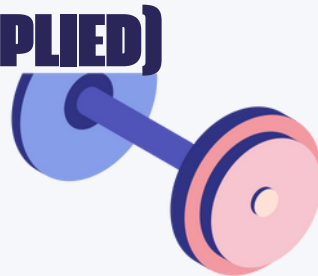
```
├ "Dumbbell Bench Press" → Contains "press" BUT NOT overhead
├   └ ⚠ RISKY - might stress shoulder
├ "Overhead Dumbbell Press" → Contains "overhead"
├   └ X REMOVED (too dangerous)
├ "Cable Fly" → No shoulder keywords
├   └ ✓ SAFE
├ "Machine Chest Press" → Contains "press"
├   └ ⚠ RISKY - rechecked manually
└ "Dumbbell Floor Press" → No shoulder keywords
  └ ✓ SAFE
```



Result: 94 safe chest exercises remain (from 144)

Removed: 50 exercises (34.7%)

SIMILARITY-BASED SELECTION (NOW WITH FILTERS APPLIED)



```
text
1. Pick first exercise from filtered pool
   └─ "Dumbbell Bench Press" (seed exercise)

2. Look up its similarity scores from matrix
   Row 127: [1.0, 0.92, 0.78, 0.65, 0.32, ...]

3. Find matches in FILTERED POOL (only the 67 safe ones)
   └─ Match 1: "Machine Chest Press" (0.92) ✓ In safe pool
   └─ Match 2: "Dumbbell Fly" (0.78) ✓ In safe pool
   └─ Match 3: "Cable Chest Press" (0.65) ✓ In safe pool
   └─ Match 4: "Smith Chest Press" (0.60) ✗ Below threshold
   └─ (Stop searching once found enough)

4. Apply similarity threshold (0.65)
   └─ Keep if similarity ≥ 0.65
   └─ Discard if similarity < 0.65
   └─ Result: Top 3-4 most similar, safe exercises
```

DIVERSITY FILTERING 'Final step' :

Problem: After similarity selection, all 4 exercises might be:

- All dumbbells(same equipment=monotonous) All
- press variations (same movement family = boring)

We handled it by family+equipment filter

WARMUP & COOLDOWN PROTOCOLS

Warmups and cooldowns are critical components of training :

- Warmup: Prepares body, raises heart rate, activates target muscles, prevents injury
- Cooldown: Aids recovery, reduces heart rate gradually, improves flexibility, prevents dizziness

our system automatically generates personalized warmups and cooldowns based on the day's muscle groups. 'predifined logic'

SYSTEM OUTPUT (demo)

YOUR PERSONALIZED WORKOUT PLAN	
PROGRAM DETAILS	
Age:	26
Gender:	Female
BMI:	23.5
Training Days:	5/week
Experience Level:	Intermediate
Fitness Goal:	WEIGHT LOSS
Training Location:	Gym
Injuries Managed:	Shoulder injury, lower back pain
DAY 1 - Chest, Triceps (7 EXERCISES)	
WARM-UP (5-7 minutes)	
1. Light Cardio - 3-5 minutes	
↳ Jumping jacks, jogging in place, or cycling	
2. Arm Circles - 30s each direction	
↳ Large circles forward/backward	
3. Band Pull-Aparts - 15 reps	
↳ Light band, squeeze shoulder blades	
MAIN WORKOUT	
1. CABLE ONE ARM FLY ON EXERCISE BALL	
Muscle Group: Chest Target: pectorals	
Equipment: cable	
Sets: 3 Reps: 12-20 Rest: 45s	
Instructions: Follow standard form	
2. CABLE ONE ARM LATERAL BENT-OVER	
Muscle Group: Chest Target: pectorals	
Equipment: cable	
Sets: 3 Reps: 12-20 Rest: 45s	
Instructions: Follow standard form	



PERSONALIZED NUTRITION (MACROS)

Algorithm: Mifflin-St Jeor Formula

- Generic calorie estimates are inaccurate. We need precision nutrition tied to the user's specific profile.
- Solution: Industry-standard Mifflin-St Jeor formula

Calculate BMR (Basal Metabolic Rate)

Formula varies by gender:

For Females:

$$\text{BMR} = 10 \times \text{weight} + 6.25 \times \text{height} - 5 \times \text{age} - 161$$

For Males:

$$\text{BMR} = 10 \times \text{weightkg} + 6.25 \times \text{heightcm} - 5 \times \text{age} + 5$$

Calculate TDEE (Total Daily Energy Expenditure)

Formula:

$$\text{TDEE} = \text{BMR} \times \text{Activity Multiplier}$$



NUTRITION RECOMMENDATIONS

Daily Calorie Targets:

- └ BMR: 1,594 kcal (metabolism at rest)
- └ TDEE: 2,471 kcal (with activity)
- └ Target: 1,971 kcal (500 deficit for weight loss)

Daily Macro Targets:

- └ Protein: 167g (30% of calories)
- └ Carbs: 287g (44% of calories)
- └ Fat: 46g (26% of calories)



SLEEP RECOMMENDATIONS



Training Volume-Based Sleep Formula

- Generic "8 hours" doesn't account for training load. Heavy training needs more recovery.
- Solution: Dynamic sleep recommendation tied to training volume, experience, and goal

Base Sleep Calculation

Formula:

Sleep Hours = $7.5 + (\text{Volume Factor}) + (\text{Experience Factor}) + (\text{Goal Factor})$

Training Volume Adjustment

Formula:

Volume Factor = $\text{Training Days} \times 0.2 \text{ hours}$

Logic: More training days = more muscle damage = need more recovery

Experience Level Adjustment

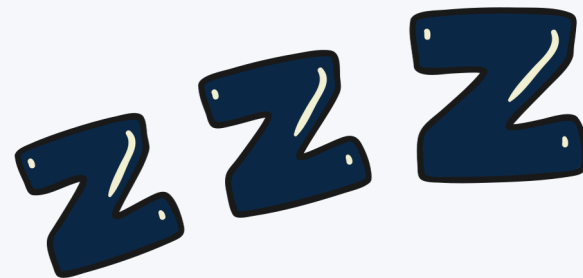
- Logic: Advanced lifters lift heavier → more damage → need more sleep

Goal-Based Adjustment

- Logic: Strength/power training depletes CNS more → needs neural recovery

Calculate Total Recommendation

- Logic: Strength/power training depletes CNS more → needs neural recovery



SLEEP RECOMMENDATIONS

text



Base Sleep: 7.5 hours
Volume Factor: 5 days $\times$ 0.2 = +1.0 hour
Experience Factor: Intermediate = +0.3 hours
Goal Factor: Weight Loss = 0 hours

TOTAL RECOMMENDED: 7.5 + 1.0 + 0.3 + 0 = 8.8 hours/night

text



SLEEP RECOVERY RECOMMENDATIONS

🔗 Target Sleep: 8.8 hours/night
(adjusted for 5-day training + intermediate recovery needs)

💡 Tips:

1. Maintain consistent sleep schedule (sleep at same time daily)
2. Avoid screens 1 hour before bed
3. Keep room cool (60-67°F / 15-19°C)
4. Stay hydrated throughout the day
5. If high volume: Prioritize sleep for recovery
6. Consider naps: 20-30 min power naps mid-day if needed

⚠️ Special Notes:

- └ Heavy training volume (5 days) requires prioritizing sleep
- └ Your arm splits need especially good recovery

Future Improvements

- The current system is production-ready with 86.81% accuracy, but we've identified strategic improvements that would further enhance personalization, safety, and user engagement. Below are prioritized enhancements based on research and user feedback patterns.

1-UNSUPERVISED PERSONALIZATION (more User-Driven Inputs)

2-Movement Family Diversity Filtering

3-INJURY HANDLING IMPROVEMENTS using NLP

4-CONTENT-BASED FILTERING IMPROVEMENTS (dynamic threshold adjustments)

5- USER EXPERIENCE ENHANCEMENTS 'feedback loops'

6-Exercise Explanation & Reasoning

7-Collaborative filtering add + Cold-Start Problem Solution

8-Nutrition meals recommendations

9-Warmups stretches dynamic (currently predefined)



Related work



611noorsaeed/Diet-and-Workout-...

Diet and Workout Recommendation system using openai

1Contributor

1Issue

22Stars

8Forks

611noorsaeed/Diet-and-Workout-Recommendation-system-using-openai: Diet and Workout Recommendation...

Diet and Workout Recommendation system using openai - 611noorsaeed/Diet-and-Workout-Recommendation-system-using-openai

 GitHub

6]:

workout_names

6]:

```
['1. Walking',  
'2. Swimming',  
'3. Yoga',  
'4. Pilates',  
'5. Strength Training',  
'6. Cycling']
```

Deployment



main Workout_Recommendationsystem 19.5 MB		Go to file		2 contributors	History: 103 commits	Contribute
hanyawael MariiHamza Update app.py (#25) 7d90282 VERIFIED about 2 hours ago						
.gitattributes	Safe	1.52 kB	Initial commit	2 days ago		
Enhanced_Fitness_Workout_Dataset.csv	Safe	969 kB	Upload 9 files	2 days ago		
README.md	Safe	383 Bytes	Initial commit	2 days ago		
app.py	Safe	46.5 kB	Update app.py (#25)	about 2 hours ago		
encoders (1).pkl	pickle	2.34 kB	Upload 9 files	2 days ago		
exercises.csv	Safe	797 kB	Upload 9 files	2 days ago		
exercises_clean.csv	Safe	1.15 MB	Upload exercises_clean.csv	2 days ago		
fitness_encoder.pkl	pickle	365 Bytes	Upload 9 files	2 days ago		
gender_encoder.pkl	pickle	259 Bytes	Upload 9 files	2 days ago		
hgb_model.joblib	SingCloud	1.53 MB	Upload hgb_model.joblib	2 days ago		
requirements.txt	Safe	98 Bytes	Update requirements.txt (#1)	1 day ago		
scaler.pkl	pickle	1.08 kB	Upload 9 files	2 days ago		
similarity_matrix (1).npy		14 MB	Upload 9 files	2 days ago		
workout_encoder.pkl	pickle	385 Bytes	Upload 9 files	2 days ago		

Performance:

- User Profile → Prediction → Exercise Recommendation: ~1 second
- Firebase async saves: <100ms
- Total response time: Fast & smooth

Status:  LIVE on Hugging Face collecting real user data for future model improvements

Deployment

Personalized Fitness Recommendation System

Complete your profile → Generate plan → Rate exercises → Submit feedback

STEP 1: Complete Your Profile

Username *

Enter username for data tracking

Basic Information

Age (years) *

25

<3

Weight (kg) *

70

<3

Height (m) *

1.7

<3

Gender *

Male

Female

Other

Fitness Profile

Fitness Goal *

Athletic Performance

Endurance

Flexibility

General Fitness

Muscle Gain

Strength

Stress Relief

Weight Loss

Experience Level *

Beginner (< 6 months)

Intermediate (6 months - 2 years)

Advanced (2+ years)

Personalized Workout Plan for HANYA

Your Profile & Health Metrics

Name	hanya	Age	25 years
Weight	70.0 kg	Height	1.6 m
Gender	Female	Experience	Intermediate
Goal	Muscle Gain	Workouts/week	4
Equipment	Gym		
Avg BPM	130	Resting BPM	70
Max BPM	195	Water Intake	3 L/day
Injuries	None		
Predicted Workout Type	FUNCTIONAL (50%)		

Your Workout Plan

Day 1: Chest + Triceps

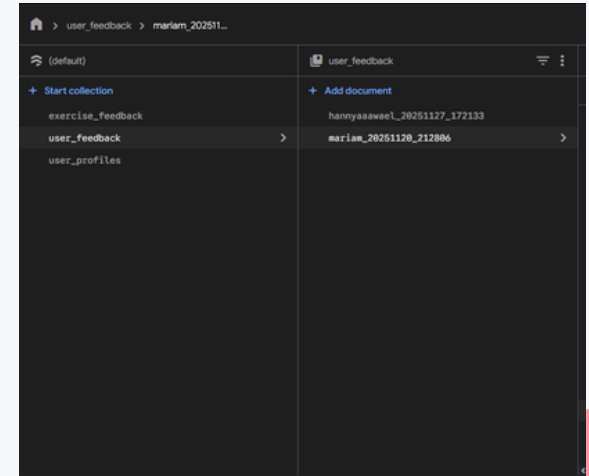
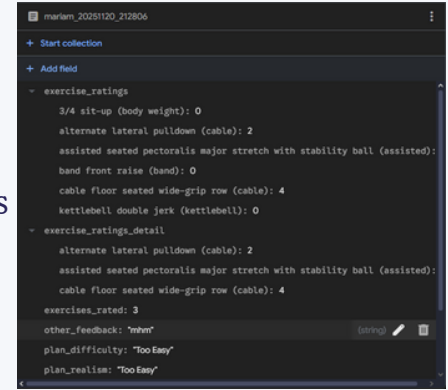
#	Exercise	Muscle	Equipment	Sets x Reps	Rest (s)	Instructions
1	anchor push up	Chest	body weight	4x8-12	60	1. Start in a push-up position with your hands slightly wider than shoulder width apart. 2. Extend one arm straight out to the side, parallel to the ground. 3. Lower your body by bending your elbows, keeping your back straight and core engaged.
2	band bench press	Chest	band	4x8-12	60	1. Lie flat on a bench with your feet flat on the ground and your back pressed against the bench. 2. Grasp the band handles with an overhand grip, slightly wider than shoulder width apart. 3. Extend your arms fully, pushing the bands away from your chest.
3	band one arm twisting chest press	Chest	band	4x8-12	60	1. Attach the band to a sturdy anchor point at chest height. 2. Stand with your side facing the anchor point and grab the band with one hand. 3. Step away from the anchor point to create tension in the band.
4	barbell bench press	Chest	barbell	4x8-12	60	1. Lie flat on a bench with your feet flat on the ground and your back pressed against the bench. 2. Grasp the barbell with an overhand grip slightly wider than shoulder width apart. 3. Lift the barbell off the rack and hold it directly above your chest with your arms fully extended.
5	band close angle push up	Triceps	band	4x8-12	60	1. Place a band around your upper arms, just above the elbows. 2. Assume a push-up position with your hands directly under your shoulders and your body in a straight line from head to heels. 3. Bend your elbows and lower your chest towards the

FEEDBACK LOOPS

Firebase is our feedback loop. Every user that rates an exercise or provides feedback feeds data back into our system. Over time, this creates a User Experience Matrix that shows us exactly which recommendations work and which don't. We use this to retrain our ML model quarterly, improving from 56% confidence today to 80%+ in 6 months. It's the difference between a static app and one that learns from every user.

Key Points:

- ✓ Captures real user data (profiles + exercise ratings + feedback)
- ✓ Builds UX metrics showing satisfaction trends
- ✓ Enables quarterly model retraining with real data
- ✓ Creates competitive advantage (system gets smarter as it grows)



TOOLS



THANK YOU!

